

DONETSK, Ukraine

WMO: 345190

Lat: 48.072N

Lon: 37.727E

Elev: 738

StdP: 14.31

Time Zone: 2.00 (EUE)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) 3.2	(d) -8.0	(e) 3.7	(f) -0.5	(g) -2.8	(h) 4.9	(i) 4.4	(j) 30.9	(k) 21.5	(l) 27.6	(m) 18.4	(n) 7.7	(o) 70	(p) 0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	92.5	66.6	88.9	66.3	85.6	65.6	70.5	83.3	68.9	81.6	67.5	80.1	10.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	100.0	74.8	64.7	94.1	73.1	63.3	89.4	72.2	34.9	83.2	33.5	81.1	32.3	80.0	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.5	(b) 22.3	(c) 19.4	(e) -8.3	(f) 95.3	(g) 6.8	(h) 4.3	(i) -13.3	(j) 98.4	(k) -17.3	(l) 100.9	(m) -21.1	(n) 103.3	(o) -26.0	(p) 106.4		
(4) 25.5	(5) 22.3	(6) 19.4	(7) -8.8	(8) 73.6	(9) 6.7	(10) 2.3	(11) -13.7	(12) 75.3	(13) -17.6	(14) 76.6	(15) -21.4	(16) 77.9	(17) -26.3	(18) 79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	24.4	25.2	35.0	49.7	60.9	67.6	72.4	70.7	60.1	48.1	36.7	27.3		
	DBStd	19.32	10.64	10.72	8.43	7.51	8.21	6.65	6.16	6.99	7.54	8.76	8.93	10.05		
	HDD50	3323	794	695	469	94	11	0	0	0	10	139	406	705		
	HDD65	6747	1259	1115	931	459	180	48	7	22	180	526	850	1170		
	CDD50	2702	0	0	4	85	348	527	695	642	313	82	6	0		
	CDD65	650	0	0	0	1	52	125	237	199	33	3	0	0		
	CDH74	6305	0	0	0	16	483	1143	2369	2026	248	20	0	0		
	CDH80	2314	0	0	0	1	113	364	953	838	41	4	0	0		
	WSAvg	9.3	10.5	11.2	11.3	9.7	8.1	7.6	7.5	7.9	8.3	8.9	9.6	10.8		
	PrecAvg	20.4	1.5	1.3	1.5	1.5	1.8	2.8	1.9	1.5	1.7	1.5	1.5	1.7		
(15) Precipitation	PrecMax	29.8	4.5	3.9	3.0	4.2	3.5	7.1	7.6	4.6	5.0	3.5	3.5	3.5		
	PrecMin	13.7	0.4	0.3	0.6	0.2	0.0	0.6	0.2	0.0	0.0	0.4	0.2	0.5		
	PrecStd	3.9	0.9	0.8	0.7	0.8	0.9	1.6	1.6	1.2	1.3	0.7	0.9	0.9		
	PrecStd	3.9	0.9	0.8	0.7	0.8	0.9	1.6	1.6	1.2	1.3	0.7	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	49.7	62.4	76.5	88.0	91.6	96.4	98.2	84.5	76.7	60.8	50.2		
		MCWB	44.5	45.7	50.1	57.7	63.5	67.1	67.4	66.8	62.5	61.0	53.2	47.8		
	2%	DB	42.7	42.9	56.2	70.0	82.7	87.7	92.8	93.1	80.3	69.6	55.4	46.1		
		MCWB	41.4	40.6	46.8	54.4	61.8	66.4	67.3	65.6	62.1	57.4	51.2	44.9		
	5%	DB	39.4	39.5	51.3	66.3	79.2	84.3	89.3	88.2	76.8	64.7	51.7	42.5		
		MCWB	38.4	37.9	43.9	52.4	60.8	65.6	67.1	65.4	60.1	56.4	48.4	41.4		
	10%	DB	36.8	37.3	46.8	62.7	75.6	80.8	85.7	84.3	73.3	60.8	48.3	39.4		
		MCWB	35.8	36.0	41.8	50.6	60.1	64.5	66.6	64.6	59.2	54.7	46.2	38.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	45.6	51.1	58.7	66.8	71.9	73.6	71.2	68.5	63.3	54.9	48.6		
		MCDB	46.3	49.2	59.9	72.9	78.0	83.8	86.4	85.8	79.6	71.3	58.2	49.6		
	2%	WB	41.3	41.0	47.9	56.1	64.7	69.1	71.1	69.1	65.3	60.1	51.6	44.5		
		MCDB	42.1	42.1	54.6	66.5	77.5	81.0	83.9	82.9	74.2	66.3	54.9	45.5		
	5%	WB	38.4	38.3	45.0	54.1	63.0	67.6	69.6	67.6	63.3	57.7	49.0	41.2		
		MCDB	39.3	39.4	49.9	63.2	75.1	79.7	82.1	80.8	71.6	63.3	51.1	42.0		
	10%	WB	35.7	36.0	42.2	52.2	61.2	66.0	68.0	66.3	61.4	55.1	46.2	38.5		
		MCDB	36.4	36.9	46.1	60.3	72.8	77.4	80.8	79.3	69.1	59.9	48.1	39.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.4	13.0	17.2	19.2	19.0	20.0	20.7	18.5	14.2	10.2	8.6		
		MCDBR	8.8	11.0	20.3	22.2	22.7	22.6	23.3	24.5	22.1	18.9	14.4	10.0		
		MCWBR	7.6	9.2	12.3	10.8	9.6	8.9	8.3	8.7	10.0	10.6	10.6	8.9		
		MCDBR	8.5	10.0	17.9	19.2	20.9	20.0	20.3	21.1	18.6	16.9	12.7	9.2		
	5% WB	MCWBR	7.7	8.8	12.0	10.4	10.3	9.2	9.1	9.1	9.9	10.9	10.2	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.356	0.391	0.449	0.412	0.412	0.437	0.435	0.407	0.373	0.348	0.331		
	taud		2.210	2.141	2.175	2.077	2.270	2.295	2.231	2.225	2.274	2.338	2.320	2.258		
	Ebn at Noon		226	246	258	253	270	270	261	256	251	240	219	209		
	Edh at Noon		26	35	39	48	41	40	42	41	35	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		305	599	939	1407	1860	1979	1961	1778	1248	733	342	215		
	RadStd		63	116	127	142	164	165	121	123	123	110	62	35		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.69	N/A	N/A	N/A	-1.17	N/A	N/A	-499	+351	+189
(47) Regional (9 neighbors)		+1.26	N/A	N/A	+2.02	N/A	N/A	N/A	-306	+212	+144

Nomenclature: See separate page